

From reach to grasp recognition to CTM decision dynamics

A plain language report of the completed P1 pilot

About 15–20 minutes including the figures. This report follows the Continuous Thought Machine (CTM) from a trained object-recognition model to a small population of internal units, then to a score pointing toward one of two objects. The main finding is that the internal state changes in an object-related way, but those changes do not yet establish a drift-diffusion process.

The strongest example is also the most informative limitation: a fixed three-unit score correctly separates all 32 selected test recordings at internal tick 15, then falls to 24 of 32 at tick 25. That peak was observed after examining the test trajectories. It is not a newly validated stopping rule or a replacement for the original classifier.

Where the trained model came from

Stage	Training and evaluation	Reported result
Pooled RTG reference	All participants' B0 → their B1–B4	85.31% on 640 recordings
Participant-specific RTG	Each person's B0 → their B1–B4	83.91% across five models
This internal analysis	Frozen P1 model; 64 B0 and 128 B1–B4 recordings	73.44% on P1 B1–B4

B0 is the training block; B1–B4 provide the changed-block evaluation. The same participants appear on both sides in these experiments. The current analysis therefore asks about transfer across blocks, not recognition in an unseen person.

The P1 model was the predetermined pilot, rather than a model chosen for especially attractive internal results. It used 75 epochs, or 150 optimizer updates. Its checkpoint, scaler and recording identities were restored, and its original predictions reproduced before activity was examined. No new model was trained for this analysis.

The path through the report

Restore the model → record internal activity → find correlated groups on B0 → check those groups on B1–B4 → visualize trajectories → define an object-directed score → compare it with the model's own decisions → decide what can reasonably be said about DDM.

Reading convention: B0 choices are fitted or discovered; B1–B4 results are evaluated with those choices held fixed. Earlier project exploration means B1–B4 is not an untouched dataset across the whole project.

What changes inside the CTM

The input is a cropped reach-to-lift movement, represented by 15 fingertip coordinates at 100 resampled positions. Scaling comes from B0. The crop follows the agreed sensor-defined endpoint and includes no extra post-lift tail.

The CTM then performs 25 internal updates, called ticks. All input positions are already available before those updates begin. A tick is a computation step, not a physical movement frame or a millisecond. This input configuration has no temporal position encoding, so the interpretation is recognition from the available hand states rather than explicit learning of their chronological order.

Signal	What it means here
256 post-activation units	The updated internal state after each recurrent step; the primary activity studied.
Synchronization	Features calculated from state activity. Action synchronization guides attention; output synchronization feeds class scores.
Logits and certainty	Eight class scores, plus certainty derived from their normalized entropy. Certainty is not the probability that the answer is correct.

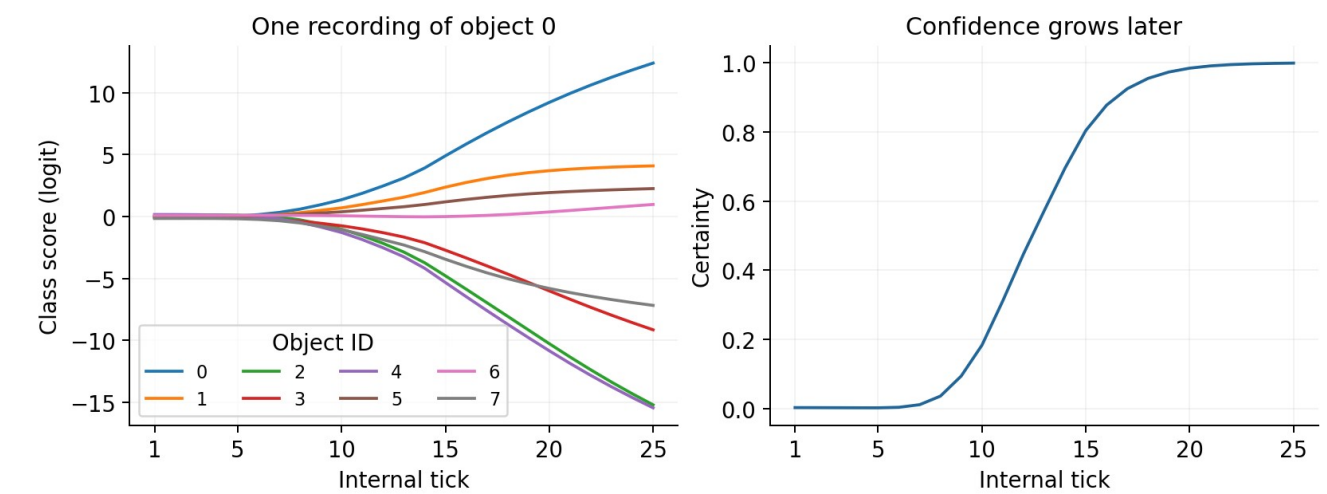


Figure 1. A deterministic P1 test example, chosen by recording order. The correct object is 0. Class preference and certainty develop on different schedules.

The extracted arrays contain $64 \times 25 \times 256$ values for B0 and $128 \times 25 \times 256$ for B1-B4. Tracking reproduced the saved outputs, and the initial learned state is not counted as an extra tick. Raw activity ranges roughly from -7 to +8; “post-activation” does not mean a value between zero and one.

For comparison plots, each unit is standardized with its own B0 mean and standard deviation. This makes small-amplitude units visible beside large-amplitude ones. The same transformation is applied to test data. It does not independently rescale each recording.

Finding groups that change together

Pearson correlation asks whether two units tend to rise and fall together. First, each unit's mean within a recording is subtracted. This removes a constant offset while keeping its changes across ticks. Correlations are then calculated across the centered recordings. Raw correlations are retained as a reference.

Complete-linkage clustering groups units only when even their least-similar pair meets the chosen cut. The primary cut is $r = 0.7$, with at least three units per candidate group. This is an exploratory rule, not a discovered biological boundary. Negative correlations are not turned into positive similarities.

Correlation cut	Candidate groups	Units in groups
0.6	19	64
0.7 primary	7	22
0.8	1	3

The threshold sensitivity matters: most units are not in the seven selected groups. “Seven groups” does not mean seven object populations or that the remaining units are unimportant.

Group	Units	B0 bootstrap coassignment	B0 mean r	Test mean r	Test min r
45	3	58.7%	.767	.824	.778
87	3	72.7%	.714	.722	.693
114	4	64.0%	.820	.809	.683
131	3	100%	.888	.706	.559
169	3	100%	.824	.845	.774
175	3	98.7%	.782	.693	.573
181	3	52.0%	.783	.770	.672

Bootstrap here means drawing whole B0 recordings again, with replacement within each object, 100 times. Coassignment measures how often original group members remain together. A value of 100% means all tested resamples, not a guarantee across new participants or model training seeds.

Group membership is frozen for B1–B4. All seven retain positive correlations, but only groups 45 and 169 keep every pair above 0.7. Removing each unit's average tick trajectory also leaves positive residual correlations in the same groups. That rules out the removed average trend as the sole explanation, not shared inputs, object responses or other common influences.

Seeing the population move

Principal component analysis (PCA) gives a compact map of the 256-unit state. It finds directions with large variation; it is not trained to maximize object accuracy. Both the activity standardization and PCA are fitted on B0, then reused unchanged on B1–B4. This makes the two plots comparable.

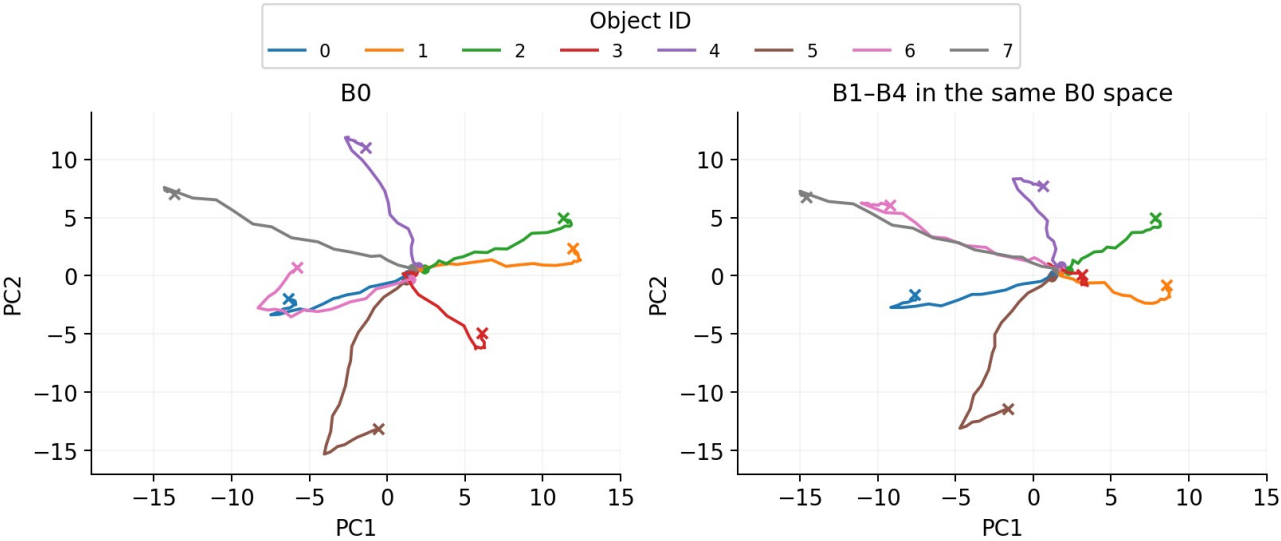


Figure 2. Object-average trajectories in shared B0 PCA coordinates. Circles mark tick 1 and crosses tick 25. Each B0 mean contains eight recordings; each test mean contains sixteen.

Projection	B0 variance represented
Whole population first two components	32.3%
Whole population first three components	42.2%
Group 131 first component	97.1%
Group 131 first two components	99.6%

The object means follow different paths, and some broad relationships remain visible under transfer. For example, several trajectories keep a similar direction while others change substantially. This is visual evidence of object-related structure; the report does not claim a quantitative test of preserved geometry.

The first two whole-population components leave about two-thirds of B0 variation outside the picture. Curves overlapping in this view can still differ elsewhere. Conversely, separated means can hide overlapping individual recordings.

Group 131 was selected for a compact follow-up using B0 bootstrap stability, with its lower group ID breaking a tie with group 169. Its units are 11, 65 and 143. A nearly one-dimensional PCA is unsurprising for strongly correlated units. It supports a shared activity direction, but does not itself identify a decision variable.

Objects and certainty are different questions

For a group readout, the standardized original activity of its member units is averaged. The temporal centering used for group discovery is not applied here: sustained offsets might also carry object information.

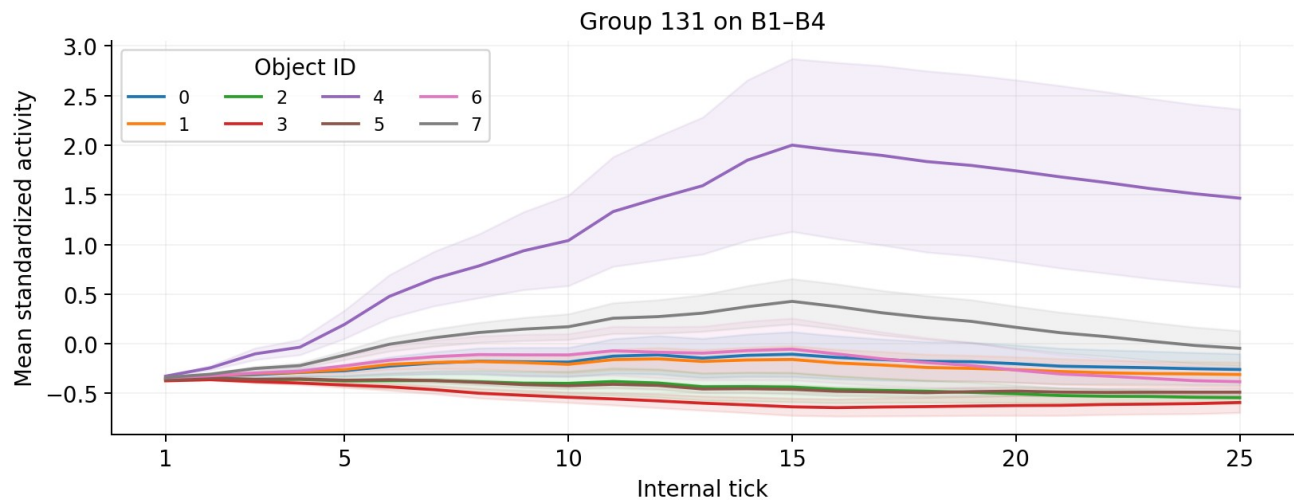


Figure 3. Group 131’s mean activity for all eight test objects, with ± 1 standard deviation across recordings. Each object has sixteen recordings. Large differences do not imply one group belongs to one object.

Group	Median object-related variance in B0	Median object-related variance in B1-B4
45	.954	.752
87	.925	.801
114	.980	.853
131	.997	.839
169	.979	.933
175	.950	.837
181	.972	.864

These values are eta squared, summarized over ticks. At each tick, it asks how much of the observed variation is associated with differences between object means. It is not decoding accuracy or a claim that the groups cause recognition. B0 values describe the model’s training recordings.

Certainty is a separate question. Correlation across ticks can be large simply because both signals change with computation. The analysis therefore also compares recordings at the same tick and, later, within the same object. Associations change with tick and object, and high certainty can accompany wrong decisions. Page 9 shows this distinction directly.

A score pointing toward two objects

To give activity a direction of choice, the analysis uses group 131’s three units. Each object gets one B0 prototype: its average standardized activity across recordings and all ticks. The farthest-apart prototypes select objects 3 and 4. This deliberately selects a favourable pilot, not a representative pair for all eight objects.

B0 prototype	Unit 11	Unit 65	Unit 143
Object 3	-0.696	-0.406	-0.424
Object 4	2.254	2.185	2.253

The distance is 4.753 in standardized coordinates. All seven largest pair distances involve object 4, so this population mainly singles out object 4 from the rest. Each selected B0 prototype uses eight recordings.

The score projects the current three-unit state onto the line between those prototypes, measured from their midpoint. Positive means toward object 3; negative means toward object 4. The direction is normalized to length one. Zero is a geometric midpoint, not a learned CTM stopping boundary. Prototypes average over all ticks, so they are not final-state prototypes.

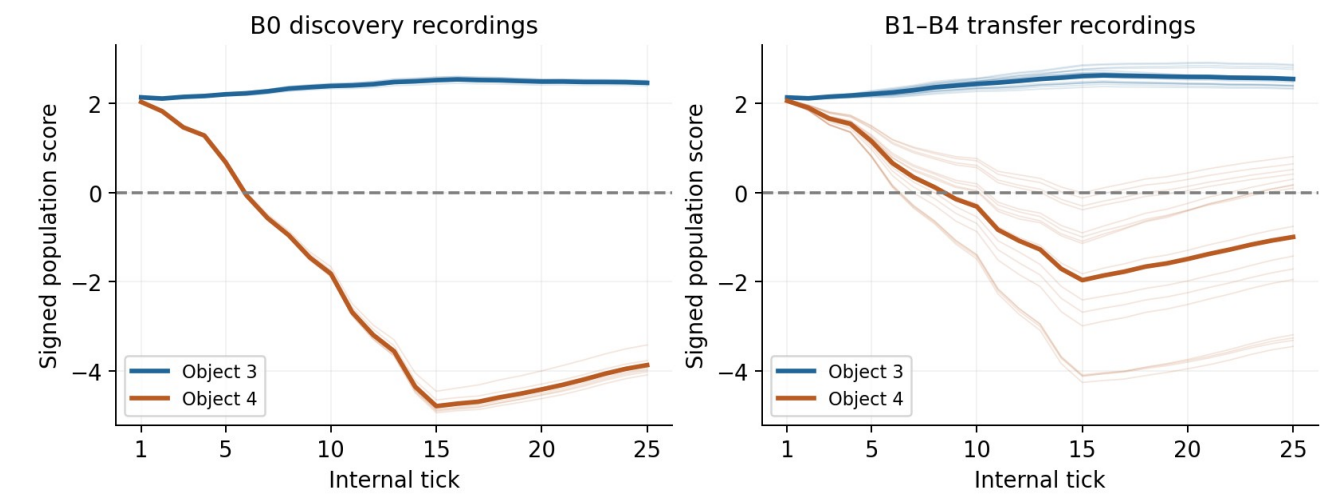


Figure 4. The same frozen score on B0 and B1–B4. Thin lines show individual recordings; thick lines show object means. The dashed line is the fixed zero cutoff. Test data contain sixteen recordings per object.

Object 3 stays positive with relatively little movement. Object 4 starts positive too, then moves negative and later partly returns. The dynamics are asymmetric: this is not a picture of two alternatives symmetrically leaving a neutral starting point. Nothing about the score direction or cutoff is refitted on test recordings.

The population score and the model disagree

There are three readouts to compare: the sign of the population score; the sign of the two selected logits' difference; and the actual eight-class prediction. The first two force a choice between objects 3 and 4. The third can choose any object.

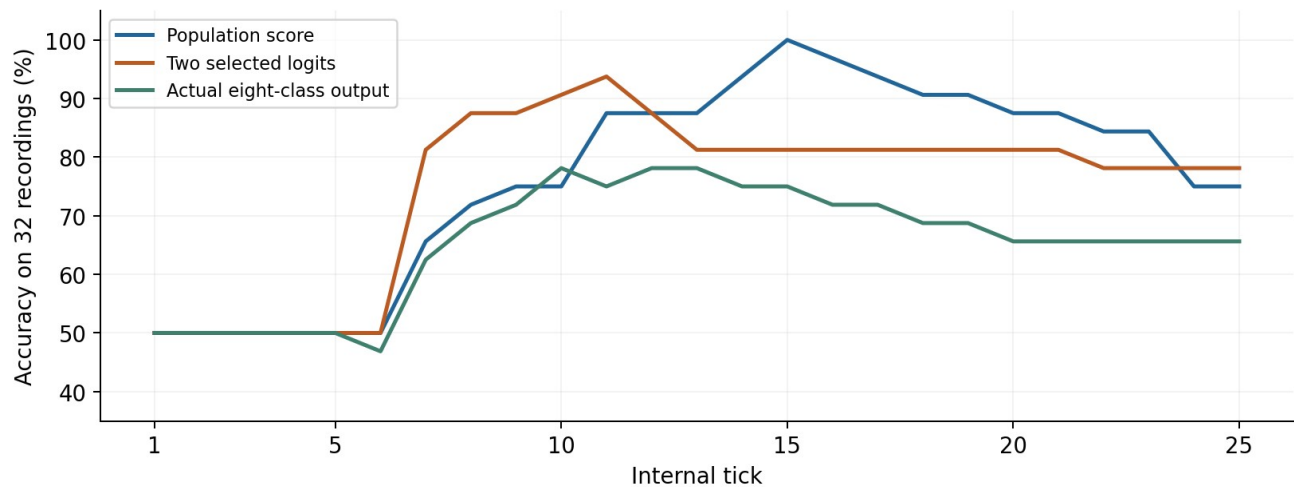


Figure 5. All three curves use the same 32 test recordings. The population score peaks at tick 15. The eight-class curve remains a different task from the forced two-object comparisons.

Tick	Population score	Two logits	Eight-class output	Third-object predictions
1	50.00%	50.00%	50.00%	0.00%
7	65.62%	81.25%	62.50%	18.75%
11	87.50%	93.75%	75.00%	25.00%
15	100.00%	81.25%	75.00%	25.00%
25	75.00%	78.12%	65.62%	34.38%

At tick 1, both binary readouts score 50%, but they make opposite choices on every recording. Their equal accuracy therefore does not mean agreement. The two-logit preference becomes useful earlier, while the population score reaches its highest accuracy later.

At tick 15, the score is correct for 32/32 recordings, compared with 26/32 for the two-logit readout and 24/32 for the actual model. At tick 25 those counts are 24/32, 25/32 and 21/32. Eleven final predictions are third objects.

The 100% score peak is an observed test-curve maximum, not independently validated performance at a chosen stopping tick. It shows that a simple population projection can contain useful information that is not identical to the model's output. This analysis does not show that changing the output layer would improve the trained model.

Why chance accuracy can hide information

At the first tick, both objects lie on the positive side of the fixed midpoint. The sign rule therefore predicts object 3 for everyone and gets 16/32 correct. Yet the two score distributions are already different: their means are about 2.126 and 2.050, with very little within-object spread.

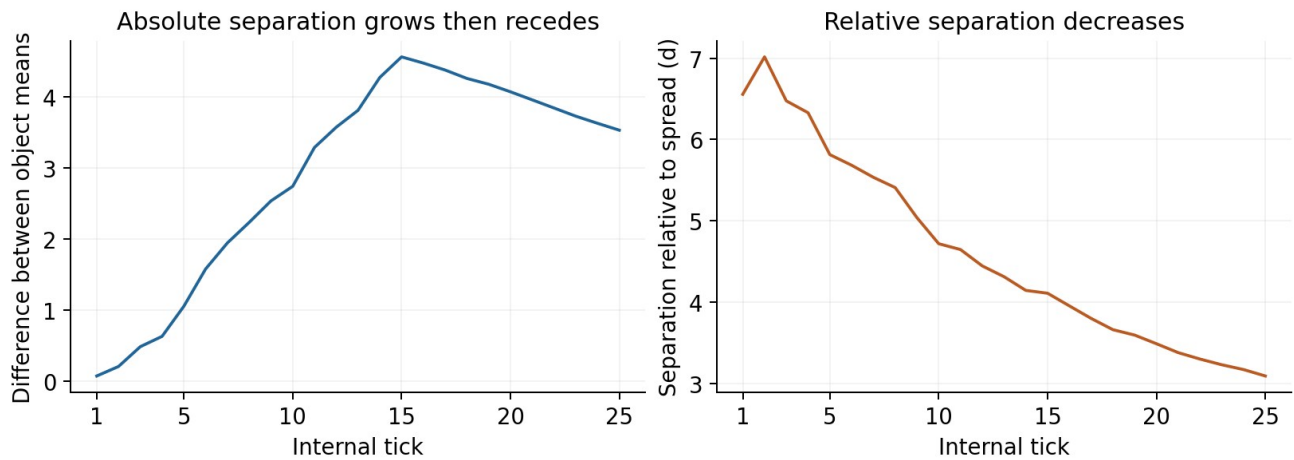


Figure 6. Left: the distance between object means. Right: that distance divided by their pooled within-object standard deviation, called Cohen's d . The two measurements answer different questions.

Tick	Difference between means	Difference relative to spread
1	0.076	6.56
15	4.566	4.11
25	3.534	3.09

The mean difference grows to tick 15, but recording-to-recording variability grows too. Relative separation therefore falls. A large d at tick 1 is not a large absolute signal and does not validate the chosen zero cutoff. We did not refit a tick-specific cutoff on test data.

True object	Original CTM outcome	Recordings	Mean zero crossings
3	Correct	13	0
3	Incorrect	3	0
4	Correct	8	1
4	Incorrect	8	2

Correctness here is the original CTM most-certain-tick decision, not correctness of the score. For object 4, incorrect cases cross toward its side and then cross back. These crossings concern the arbitrary midpoint; they are not measured absorbing-boundary decisions. This is a useful association with errors, not proof of their cause.

How dependable is this small pilot

The pair-selection check resamples whole B0 recordings within each object and recalculates the prototypes. It keeps the original group and B0 standardization fixed, so it measures conditional stability of the pair and direction, not uncertainty in the whole model or pipeline.

Check over 100 resamples	Result
Pair 3 versus 4 selected	100 out of 100
Mean direction similarity to original	0.999993
Lowest direction similarity	0.999961

Direction similarity is the cosine between two directions: one means identical orientation. The score definition is very stable under this particular check. That does not establish transfer to another participant, training seed or object pair.

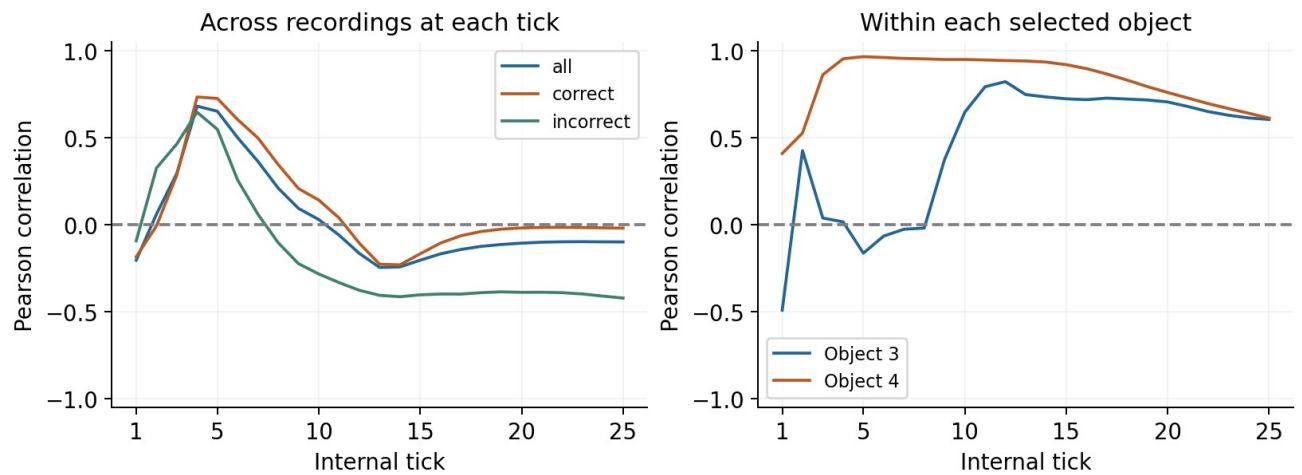


Figure 7. Group 131 activity versus certainty on test data. Left compares recordings at fixed ticks, split by the original CTM outcome. Right holds the true object fixed, with sixteen recordings per object. This uses group-mean activity, not the signed score.

The left plot changes sign during computation. The right shows why object composition matters: activity and certainty relate differently within objects 3 and 4. Pooling them can conceal or reverse associations. The sign of the signed population score also depends on the arbitrary orientation chosen for its axis, so it should not be confused with “more activity means more confidence.”

These are descriptive correlations. The 25 internal ticks are repeated states of a recording, not 25 independent participants. The pilot has one model, one participant, eight training recordings per selected object and sixteen test recordings per selected object. The two-object accuracy curve moves in steps of 1/32, or 3.125 percentage points.

What this says about DDM

A conventional two-choice drift-diffusion model describes noisy evidence moving toward a decision boundary, linking choices with when a decision is made. Our experiment records a fixed 25-tick computation. It does not observe a CTM stopping event or a behavioural reaction time, and it has not fitted or compared diffusion models [1].

The population signal is informative, but it is not yet an identified evidence accumulator. Object information is present at the first tracked update. The zero cutoff is hand-defined geometry; the later peak and reversal depend on that readout. The model continues computing after crossing it.

A necessary qualification: noisy diffusion paths can reverse before reaching a boundary, and starting biases or asymmetry can be modelled. Reversals, early information and asymmetric trajectories do not by themselves disprove DDM. The defensible decision is narrower: this work has not established the measurements and model comparisons needed for a quantitative DDM interpretation.

What the work establishes

A restored RTG model contains small correlated groups with partly transferable relationships. Those groups carry object-related activity. A B0-defined three-unit direction distinguishes a selected test pair, but its usefulness changes across ticks and differs from the eight-class output. This gives a concrete view of internal decision dynamics rather than a fitted theory of decision-making.

What remains open

The finding is limited to a favourable P1 pilot. There is no causal unit intervention, independent validation of a stopping tick, or diffusion-versus-alternative model comparison. A later study could define a defensible stopping rule, vary available evidence, and compare diffusion with a simpler settling or transformation model. Those are future questions, not missing results silently assumed here.

A short version to explain aloud

“I first checked that I was looking inside the same trained RTG model. I found small groups of units that change together and checked them on later blocks. I then turned one group into a score pointing toward one of two objects. That score became useful, peaked, and sometimes reversed, while the model’s own output followed a different schedule. So I can describe structured internal decision dynamics, but I have not shown that the CTM implements a drift-diffusion process.”

Evidence and reading note

Source: 04_ctm_internal_dynamics_ddm.ipynb, 156 cells, snapshot reviewed on 30 September 2026. Foundation exports and p1_population_analysis outputs supply the results. The P1 checkpoint is the 20260928T141738Z participant-specific B0-to-B1-B4 run. RTG context comes from the previously verified final pooled and participant-specific experiments. Figures in this report are redrawn from the saved arrays; no model was retrained. Repetitive diagnostics and code are omitted.

[1] Ratcliff and McKoon (2008), The Diffusion Decision Model: Theory and Data for Two-Choice Decision Tasks.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC2474742/> — background for the DDM interpretation, not evidence for the CTM results.